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Work Package 8

Methodology

Deliverable 8.1

Literature overview

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1.30. Gang-Hoon, K., Trimi, S., & Ji-Hyong, C. (2014). Big-Data Applications in the Government Sector. Communications Of The ACM, 57(3). 23

1.31. Beyer M. A., Thoo E., Selvage M. Y. and Zaidi E. (2017): Magic Quadrant for Data Integration Tools. Gartner report, August version 24

1.32. Singh D. and Reddy C.K. (2015): A Survey on Platforms for Big Data Analytics. Journal of Big Data, Springer 2015. 24

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1. Introduction

The aim of the literature review is to classify and show papers that are relevant according to the apply of Big Data in official statistics. The variety of the papers included in this deliverable allows using them in different statistical domains with various data sources.

The main part of the deliverable is a relevant literature section. It includes a short characteristics of each paper with the respect to the following categories: data soruces, domains, keywords. Each paper has a full bibliographic data and a link (if possible) that the reader interested in the paper may have a quick access to the full content. Some of the papers are restricted and not possible to access without purchasing subscription. We believe, that official statistics have a subscription to most of the digital libraries and it will not be an issue to access the paper.

The target audience of the paper is official statistics employers interested in developing Big Data project or any data scientists who are looking for new opportunities for extending its projects.

The final revelant literature will be published online with the search engine that allows filtering them based on the data sources, domains and keywords.

2. Relevant literature

1.1. AAPOR (2013): Report of the Task Force on Non-probability sampling, June

SPECIFICATION	DESCRIPTION
Bibliographic data	AAPOR (2013): Report of the Task Force on Non-probability sampling, June.
Link	https://www.aapor.org/AAPOR_Main/media/MainSiteFiles/NPS_TF_Report_Final_7_revised_FNL_6_22_13.pdf
Short overview (strengths, weaknesses)	The report shows different aspects on non-probability sampling, including sampling matching, network sampling, estimation and weight adjustment methods or measures of quality. Therefore, the paper is a good start to work on data sources with non-probability sampling. However the paper does not provide a complete framework to prepare non-probability sampling, it is rather a discussion on the topic.
Data sources	Social networks, surveys
Domains	Population
Keywords	Non-probability sampling, data quality
Classification (A – very relevant, B	C

– relevant, C – less relevant)	

1.2. AAPOR (2015): American Association for Opinion Research Report on Big Data

SPECIFICATION	DESCRIPTION
Bibliographic data	AAPOR (2015): American Association for Opinion Research Report on Big Data
Link	http://www.aapor.org/Education-Resources/Reports/Big-Data.aspx
Short overview (strengths, weaknesses)	The paper is a discussion on what Big Data is and how to apply Big Data methods for official statistics. Three cases are shown for online prices, social media messages and traffic with infrastructure. The cases are very briefly described. Five challenges are also discussed: data ownership, stewardship, data collection authority, privacy and reidentification, meaning of "reasonable means". The paper is just a discussion, no concrete methods to apply. Just inspiration for using Big Data.
Data sources	Social media, road sensors, web data
Domains	Economy and Finance, Population, Tourism
Keywords	Social media, price index
Classification (A – very relevant, B – relevant, C – less relevant)	C

1.3. Arai, Z. Fan, D. Matekenya & R. Shibasaki (2016): Comparative Perspective of Human Behavior Patterns to Uncover Ownership Bias among Mobile Phone Users

SPECIFICATION	DESCRIPTION
Bibliographic data	A. Arai, Z. Fan, D. Matekenya & R. Shibasaki (2016): Comparative Perspective of Human Behavior Patterns to Uncover Ownership Bias among Mobile Phone Users
Link	http://www.mdpi.com/2220-9964/5/6/85
Short overview (strengths, weaknesses)	The paper shows how the possibility of using CDR for official statistics by "analyzing routines based on time spent at key locations and compare these data with those of the general population".

	The paper shows the results on Dhaka population. The challenge for most of countries is to have access to CDR (Call Detail Records)/MCR (Mobile Call Records), which is a key issue to apply the same case. There is no detailed information on how to apply the similar case, the paper is just an inspiration to start similar projects.
Data sources	MCR (Mobile Call Records), CDR (Call Detail Records)
Domains	Tourism, Population (human behaviour)
Keywords	Call detail records, Mobile call records
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.4. Assay, M. (2012): Big Data is now Too Big - and we're drowning in toxic information, Just why are we hoarding every last binary bit?, The Register, Cloud Business

SPECIFICATION	DESCRIPTION
Bibliographic data	M. Assay (2012): Big Data is now TOO BIG - and we're drowning in toxic information, Just why are we hoarding every last binary bit?, The Register, Cloud Business, 4 June
Link	http://www.theregister.co.uk/2012/06/04/big_data_too_big/
Short overview (strengths, weaknesses)	The paper is a discussion with various references to notable comments regarding Big Data. The paper emphasizes the problem of noise in the data, however there are no solutions on how to avoid this. Therefore, the usability of the paper is for the discussion on the importance of the problems with Big Data and results of this.
Data sources	Stock prices, inflation (only references, no detailed specification)
Domains	Population, Economy and Finance
Keywords	Data noise, data quality
Classification (A – very relevant, B – relevant, C – less relevant)	C

1.5. Beyer, M.A. and L. Douglas (2012): The Importance of Big Data: A Definition. Gartner report, June version, ID Number: G00235055

SPECIFICATION	DESCRIPTION
Bibliographic data	Beyer, M.A. and L. Douglas (2012): The Importance of Big Data: A Definition. Gartner report, June version, ID Number: G00235055.
Link	https://www.gartner.com/doc/2057415/importance-big-data-definition
Short overview (strengths, weaknesses)	The most important paper regarding the definition of Big Data create by Gartner group. The paper discusses different Big Data issues – necessary changes, technological aspects, quality, skills. Although the paper does not provide any solution for Big Data work, it is worth reading as it is authorized by the company that created the common definition of Big Data.
Data sources	Not specified
Domains	Not specified
Keywords	Data quality, Big Data definition, technology
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.6. Biemer, P. (2014): Total Survey Error: Adapting the Paradigm for Big Data

SPECIFICATION	DESCRIPTION
Bibliographic data	Biemer, P. (2014): Total Survey Error: Adapting the Paradigm for Big Data
Link	https://www.niss.org/sites/default/files/biemer_ITSEW2014_Presentation.pdf
Short overview (strengths, weaknesses)	The presentation shows a framework for a total survey error – its definition and concept. In the first slides it shows how the dataset is constructed (Variables, Units). Then it proposes and discusses the Big Data processing framework which is relevant when creating ETL (Extraction, Transformation, Loading) processes.
Data sources	Not specified
Domains	Not specified

Keywords	ETL (Extraction, Transformation, Loading), big data processing, total survey error
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.7. Braun, M. (2015): Three Things About Data Science You Won't Find In the Books. Weblog

SPECIFICATION	DESCRIPTION
Bibliographic data	Braun, M. (2015): Three Things About Data Science You Won't Find In the Books. Weblog 5th April.
Link	https://dzone.com/articles/three-things-about-data
Short overview (strengths, weaknesses)	The paper discusses the need and issues regarding evaluation, model selection and feature extraction. It concentrates on issues such as machine learning and training dataset creation. Different types of machine learning are discussed – linear models and SVM (Support Vector Machines). The paper is just a discussion – it is worth reading on how to avoid common issues on Big Data applications.
Data sources	Not specified
Domains	Not specified
Keywords	Machine learning, data quality
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.8. Chandramohan, A., Mylaraswamy D., Brian Xu, Dietrich P. (2014): Big Data Infrastructure for Aviation Data Analytics

SPECIFICATION	DESCRIPTION
Bibliographic data	Chandramohan, A., Mylaraswamy D., Brian Xu, Dietrich P. (2014): Big Data Infrastructure for Aviation Data Analytics
Link	http://ieeexplore.ieee.org/document/7015483/
Short overview (strengths, weaknesses)	The paper describes the development and adoption by honeywell Aerospace of a big data infrastructure for analyzing aviation data.

	Data is collected from sensors covering more than 1000 flight parameters with some reports claiming half a terabyte of data per flight. This data is combined with data from ACMS (Aircraft condition monitoring systems) and various repair databases to find outliers and hidden trends. The use case is using advanced econometrics like entropy analysis. The paper shows how integrating the data can assist the predictive maintenance of parts.
Data sources	Flight Sensors
Domains	Sensor generated data
Keywords	Data Visualization, Data combining, Data Mining
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.9. Choi, H., Varian H. (2011): Predicting the present with Google Trends, Technical Report

SPECIFICATION	DESCRIPTION
Bibliographic data	Choi H, Varian H. (2011): Predicting the present with Google Trends, Technical Report
Link	http://people.ischool.berkeley.edu/~hal/Papers/2011/ptp.pdf
Short overview (strengths, weaknesses)	The papers shows how to forecast economic indicators with Google Trends. It includes automobile sales, unemployment claims, travel destination planning, and consumer confidence. The examples are in R language. This paper shows very original approach on how to analyze Google Trends data. The framework can be applied to various Big Data projects, not limited to the examples used by authors.
Data sources	Google Trends, Web searches data
Domains	Economy and Finance, Tourism, Social statistics
Keywords	Unemployment, automobile sales, travel destinations, consumer confidence, Google Trends
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.10. Daas, P.J.H., M.J.H. Puts, Social Media Sentiment and Consumer Confidence, Statistics Paper Series No 5 / September 2014, European Central Bank

SPECIFICATION	DESCRIPTION
Bibliographic data	Social Media Sentiment and Consumer Confidence: Piet J.H. Daas and Marco J.H. Puts, Statistics Paper Series No 5 / September 2014, European Central Bank
Link	https://www.ecb.europa.eu/pub/pdf/scpsps/ecbsp5.en.pdf
Short overview (strengths, weaknesses)	Facebook (10%), Twitter (80%), Linkedin messages etc. were collected from external company. Data was analyzed with R based on CSV (Comma Separated Values) files. The paper has a detailed description of the method used (e.g., Pearson, regression) to provide the results related to consumer confidence.
Data sources	Social media – Facebook, Twitter, Linkedin
Domains	Social statistics, Population
Keywords	Social media, sentiment analysis
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.11. Daas, P.J.H., Puts, M.J.H. (2014): Sentiment analysis of Mexican tweets: smileys and emoticons. A Big Data sandbox studies for the social data task team of the UNECE taskforce, UNECE

SPECIFICATION	DESCRIPTION
Bibliographic data	Daas, P.J.H., Puts, M.J.H. (2014): Sentiment analysis of Mexican tweets: smileys and emoticons. A Big Data sandbox studies for the social data task team of the UNECE taskforce, UNECE.
Link	https://statswiki.unece.org/display/bigdata/Experiment+report:++Social+Media+-+Sentiment+Analysis#b88ce60eff8645719dabf855ebaf812e
Short overview (strengths, weaknesses)	Good paper to start with sentiment analysis on tweets. The paper shows an analysis of the 92 million tweets. It gives a brief description on the Twitter API and tools used.

	The process was divided into three phases: data acquisition, processing and output. The Mexican Consumer Confidence is the main contribution in the paper.
Data sources	Twitter (social media)
Domains	Social statistics (sentiment analysis), Population
Keywords	Twitter, Sentiment analysis, Social media
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.12. Demunter, C. , G. Seynaeve (2017): Better quality of mobile phone data based statistics through the use of signalling information – the case of tourism statistics, NTTS Conference

SPECIFICATION	DESCRIPTION
Bibliographic data	C. Demunter, G. Seynaeve (2017): Better quality of mobile phone data based statistics through the use of signalling information – the case of tourism statistics, NTTS Conference, 13-17 March 2017 (paper and presentation download page)
Link	http://nt17.pg2.at/data/abstracts/abstract_160.html?zoom_highlight
Short overview (strengths, weaknesses)	Authors are using mobile phone data from one of Belgian operators. They show a detailed analysis of the data, including biases, extrapolations, continuity etc. The data shows outbound trips of mobile phone holders, also by destination country. The strength of the paper is a set of charts with mobile phone users distribution regarding different tourism issues. It can be an inspiration on how to use mobile phone data for tourism statistics.
Data sources	Mobile phone data, signaling data
Domains	Tourism
Keywords	Signaling data, mobile phone data
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.13. Glasson, M., J. Trepanier, V. Patruno, P. Daas, M. Skaliotis and A, Khan (2013): What does Big Data mean for Official Statistics? Paper for the High-Level Group for the Modernization of Statistical Production and Services

SPECIFICATION	DESCRIPTION
Bibliographic data	Glasson, M., J. Trepanier, V. Patruno, P. Daas, M. Skaliotis and A, Khan (2013): What does Big Data mean for Official Statistics? Paper for the High-Level Group for the Modernization of Statistical Production and Services.
Link	https://statswiki.unece.org/pages/viewpage.action?pageId=77170614
Short overview (strengths, weaknesses)	This paper shows different aspects of Big Data - including data sources and issues, such as legislative, privacy, financial etc. The paper does not show any aspects regarding how to apply Big Data projects. It is worth reading to know what issues we have to tackle with when dealing with Big Data, but the paper is very general.
Data sources	Administrative data, sensors, mobile phones data, online searches, social media, credit cards
Domains	Social statistics, Economy and Finance, Tourism
Keywords	Big Data definition, Big Data issues, Big Data examples
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.14. Hajoui, O., Talea M., Bakhouyi A., Batouta Z. Dehbi R. (2016): A comparative analysis of differente approaches for Big Data Interoperability

SPECIFICATION	DESCRIPTION
Bibliographic data	Hajoui O., Talea M., Bakhouyi A., Batouta Z. Dehbi R. (2016): A comparative analysis of differente approaches for Big Data Interoperability
Link	http://ieeexplore.ieee.org/document/7831345/
Short overview (strengths, weaknesses)	The paper proposes a literature review of papers related to big data interoperability approaches. As the NoSQL databases although effective for one specific source of big data can span heterogeneous databases, models, implementations and languages this making it difficult to ensure data migration across platforms, data interoperability and data integration.

	The paper also uses a multi-criteria analysis to compare different interoperability approaches.
Data sources	Diverse
Domains	Diverse
Keywords	Data interoperability, Data Integration
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.15. Heerschap, N.M., Ortega Azurduy, S.A., Priem, A.H. and Offermans, M.P.W. (2014): Innovation of tourism statistics through the use of new Big Data sources, paper presented at the Global Forum on Tourism Statistics, Prague

SPECIFICATION	DESCRIPTION
Bibliographic data	Heerschap, N.M., Ortega Azurduy, S.A., Priem, A.H. and Offermans, M.P.W. (2014): Innovation of tourism statistics through the use of new Big Data sources, paper presented at the Global Forum on Tourism Statistics, Prague.
Link	http://www.tsf2014prague.cz/assets/downloads/Paper%201.2_Nicolaes%20Heerschap_NL.pdf
Short overview (strengths, weaknesses)	The paper presents an overview of the use of Big Data for tourism statistics. It includes specifically the use of internet robots for collecting information on tourism accommodations. The data presented in the paper are fictitious. The second case shows the possibility of the use of mobile phone records, e.g.,: 1) case for Netherlands on to show travel behavior, 2) roaming service for Portuguese tourists in Netherlands. The paper itself is an inspiration, not giving a solution.
Data sources	Websites, mobile phone records
Domains	Tourism statistics
Keywords	Mobile phone records
Classification (A – very relevant, B – relevant, C – less relevant)	B (lack of detailed methodological notes)

1.16. Jonas, J. (2012): Interview: Data protection challenge of the future: Big Data. Data Protection Law and Policy Newsletter 9(7)

SPECIFICATION	DESCRIPTION
Bibliographic data	Jonas, J. (2012): Interview: Data protection challenge of the future: Big Data. Data Protection Law and Policy Newsletter 9(7)
Link	http://www.e-comlaw.com/data-protection-law-and-policy/hottopics_template.asp?id=Jonas
Short overview (strengths, weaknesses)	A short paper on how Big Data can be defined and what are the implications when using Big Data. Just general information what are benefits when using Big Data. For instance, for technological aspects only few examples are listed. The paper can be treated as an introduction to Big Data issues.
Data sources	Social media, Google flu trends – just mentioned in text, not specified at all
Domains	Economy and Finance, Health
Keywords	Big Data benefits, Big Data implications
Classification (A – very relevant, B – relevant, C – less relevant)	C

1.17. Marchetti, S., Giusti, C., Pratesi, M., Salvati, N., Giannotti, F., Pedreschi, D., Rinzivillo, S., Pappalardo, L. , Gabrielli, L. (2015): Small area model-based estimators using big data sources. Journal of Official Statistics 31(2)

SPECIFICATION	DESCRIPTION
Bibliographic data	Marchetti, S., Giusti, C., Pratesi, M., Salvati, N., Giannotti, F., Pedreschi, D., Rinzivillo, S., Pappalardo, L. , Gabrielli, L. (2015): Small area model-based estimators using big data sources. Journal of Official Statistics 31(2), 263–281.
Link	https://www.degruyter.com/view/j/jos.2015.31.issue-2/jos-2015-0017/jos-2015-0017.xml
Short overview (strengths, weaknesses)	The paper shows the use of Big Data for small area estimation. Very detailed description with examples and equations shows indicators such as at-risk-of-poverty rate. It also shows the possibilities of the use of Big Data in covariates in area-level small area models.

	Therefore, the paper can be classified as must read for statisticians who want to use Big Data for small area estimates.
Data sources	EU-SILC (European Union Statistics on Income and Living Conditions), Mobility data
Domains	Social statistics
Keywords	Data mining, social mining, small area statistics
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.18. Maślankowski, J. (2014) Data Quality Issues Concerning Statistical Data Gathering Supported by Big Data Technology. In: Communications in Computer and Information Science, vol 424.

SPECIFICATION	DESCRIPTION
Bibliographic data	Maślankowski, J. (2014) Data Quality Issues Concerning Statistical Data Gathering Supported by Big Data Technology. In: Communications in Computer and Information Science, vol 424. Springer.
Link	https://link.springer.com/chapter/10.1007/978-3-319-06932-6_10
Short overview (strengths, weaknesses)	The paper is an overview with examples regarding the quality issues when collecting statistical data. The framework is based on the well-known statistical frameworks and is a brief overview of the data quality. The paper shows how data quality can be measured but not exploring them in a detailed way.
Data sources	Web data, traditional surveys
Domains	Not specified
Keywords	Data quality, Data comparison
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.19. Maślankowski, J. (2016) Towards De-duplication Framework in Big Data Analysis. A Case Study. In: Lecture Notes in Business Information Processing, vol 264. Springer.

SPECIFICATION	DESCRIPTION

Bibliographic data	Maślankowski, J. (2016) Towards De-duplication Framework in Big Data Analysis. A Case Study. In: Lecture Notes in Business Information Processing, vol 264. Springer.
Link	https://link.springer.com/chapter/10.1007/978-3-319-46642-2_7
Short overview (strengths, weaknesses)	The paper concentrates on two most crucial quality issues such as ambiguousness and duplicates. From this perspective it shows how duplicates are created in the dataset and how MapReduce framework is vulnerable to this. The conceptual framework is a beginning work on how to eliminate duplicates in the dataset. The paper shows three cases how to handle issues with duplicates but it is not the complete solution on this.
Data sources	Web data
Domains	Economy and finance
Keywords	Data quality, Data comparison, Real-estate market
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.20. SportLaw (2012): Socialympics: How Sports Organizations and Athletes used Social Media at London 2012

SPECIFICATION	DESCRIPTION
Bibliographic data	SportLaw (2012): Socialympics: How Sports Organizations and Athletes used Social Media at London 2012.
Link	http://www.sportlaw.ca/wp-content/uploads/2013/01/Social-Media-and-the-Games.pdf
Short overview (strengths, weaknesses)	The paper shows how Facebook and Twitter can be used to provide information on how it is used in sport domain. It shows just the results, so it may be an inspiration on what indicators can be provided based on the social media. It does not include any specific and detailed information on how to use them.
Data sources	Social Media
Domains	Social statistics, Sport
Keywords	Social media, sport statistics
Classification (A – very relevant, B – relevant, C – less relevant)	C

1.21. Taylor L. , Schroeder R, Meyer E (2014): Emerging practices and perspectives on Big Data analysis in economics: bigger and better or more of the same

SPECIFICATION	DESCRIPTION
Bibliographic data	Taylor L. , Schroeder R, Meyer E (2014): Emerging practices and perspectives on Big Data analysis in economics: bigger and better or more of the same
Link	http://journals.sagepub.com/doi/pdf/10.1177/2053951714536877
Short overview (strengths, weaknesses)	Although the terminology of Big Data has so far gained little traction in economics, the availability of unprecedentedly rich datasets and the need for new approaches – both epistemological and computational – to deal with them is an emerging issue for the discipline. Using interviews conducted with a cross-section of economists, this paper examines perspectives on Big Data across the discipline, the new types of data being used by researchers on economic issues, and the range of responses to this opportunity amongst economists.
Data sources	Semi-structured interviews with economists
Domains	Business statistics, Economy and Finance
Keywords	Big data, economics, econometrics, epistemology, business
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.22. Tennekes, M., E. de Jonge and P. Daas (2013): Visualizing and Inspecting Large Datasets with Tableplots. Journal of Data Science 11

SPECIFICATION	DESCRIPTION
Bibliographic data	Tennekes, M., E. de Jonge and P. Daas (2013): Visualizing and Inspecting Large Datasets with Tableplots. Journal of Data Science 11, 43-58.
Link	http://www.jds-online.com/file_download/379/JDS-1108.pdf
Short overview (strengths, weaknesses)	The paper shows examples in how to use R with visualization techniques. It explains how to read and interpret such data. It shows very detailed information on the technology used (R, packages etc.). All statisticians interested in alternative ways of data visualization must read the paper.

Data sources	Census, Business Survey Data (Structural Business Statistics Survey)
Domains	Economy and Finance, Population, Social statistics
Keywords	Data visualization, Census data, Survey data
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.23. Tromp, E. (2011): Multilingual Sentiment Analysis on Social Media. Master thesis, TU Eindhoven, July 16

SPECIFICATION	DESCRIPTION
Bibliographic data	Tromp, E. (2011): Multilingual Sentiment Analysis on Social Media. Master thesis, TU Eindhoven, July 16
Link	http://www.win.tue.nl/~mpechen/projects/pdfs/Tromp2011.pdf
Short overview (strengths, weaknesses)	The dissertation shows very detailed information on the framework and theoretical aspects regarding multilingual sentiment analysis on social media. As it shows very specific information regarding the use of the framework, including equations used to calculate different values, it can be easily applied by statisticians to explore multilingual sentiments. Also very important is that the dissertation includes the detailed description of the datasets collected, e.g., the list of variables from different datasets. It allows to specify what relevant variables can be collected from social media.
Data sources	Social media
Domains	Social statistics
Keywords	Sentiment analysis, social media
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.24. UN Global Pulse (2015): Analysing Social Media to understand Public Perceptions of Sanitation

SPECIFICATION	DESCRIPTION
Bibliographic data	UN Global Pulse (2015): Sanitation_2014_0.pdf Analysing Social Media to understand Public Perceptions of Sanitation
Link	http://www.unglobalpulse.org/sites/default/files/UNGP_ProjectSeries_Perceptions_Sanitation_2014_0.pdf
Short overview (strengths, weaknesses)	A short paper shows how social media can be used for water supply and sanitation issues. It shows the analysis of 260 thous. tweets related to sanitation. The paper can help to understand how to make analysis of the specific topics discussed in social media. Two figures presents the phenomenon of sanitation issues.
Data sources	Social media
Domains	Social statistics
Keywords	Sanitation, water supply, social media
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.25. Wesolowski, A., N. Eagle, A. M. Noor, R. W. Snow & C. O. Buckee (2013): The impact of biases in mobile phone ownership on estimates of human mobility

SPECIFICATION	DESCRIPTION
Bibliographic data	Wesolowski A., N. Eagle, A. M. Noor, R. W. Snow & C. O. Buckee (2013): The impact of biases in mobile phone ownership on estimates of human mobility
Link	http://rsif.royalsocietypublishing.org/content/10/81/20120986
Short overview (strengths, weaknesses)	The paper shows population movement in Kenya over a year by mobile phone data (15 million records). It also shows the data combining of the results from survey of socio-economic status to provide more information by analysis of CDR (Call Detail Records). The use case is using advanced econometrics like entropy analysis. The paper shows good practices on how to use CDR (Call Detail Records) data with advanced data analysis to provide statistical data.
Data sources	MCR (Mobile Call Records), CDR (Call Detail Records)
Domains	Tourism, Population

Keywords	Mobile call records, Data combining, Entropy
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.26. Kim, G., and Chambers, R. (2012): Regression analysis under incomplete linkage. Statistica Neerlandica, 56(9).

SPECIFICATION	DESCRIPTION
Bibliographic data	Kim, G., and Chambers, R. (2012): Regression analysis under incomplete linkage. Statistica Neerlandica, 56(9), pp. 2756–2770.
Link	http://www.sciencedirect.com/science/article/pii/S0167947312001089
Short overview (strengths, weaknesses)	Important paper for the data quality regarding the data linkage quality indicator. The goal of the paper is to show how the data in two different data sets can be linked without biases. It concerns different types of data, however the data linkage can be used to: merge of large databases, eliminate of duplicates or combine data from surveys and registers.
Data sources	Registers (total population), surveys (sample frame)
Domains	All regarding registers and surveys
Keywords	Data combining, Data quality
Classification (A – very relevant, B – relevant, C – less relevant)	A

1.27. Lavallée, P. (2015): Sample matching: Toward a probabilistic approach for web surveys and big data?

SPECIFICATION	DESCRIPTION
Bibliographic data	Lavallée, P. (2015): Sample matching: Toward a probabilistic approach for web surveys and big data?
Link	http://www.fields.utoronto.ca/video-archive/static/2015/04/320-4530/mergedvideo.ogv
Short overview (strengths, weaknesses)	The paper shows how the sample data and big data can be matched. It is called sampling matching. It is mostly on data combining however we can see that

Data sources	Web panels, Sample surveys
Domains	Social statistics
Keywords	Sample matching, Data combining
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.28. Rivers, D. (2007): Sampling for web surveys. In Joint statistical meeting

SPECIFICATION	DESCRIPTION
Bibliographic data	Rivers, D. (2007): Sampling for web surveys. In Joint statistical meeting.
Link	http://s3.amazonaws.com/static.texastribune.org/media/documents/Rivers_matching4.pdf
Short overview (strengths, weaknesses)	The paper shows how sample matching can be treated on collecting the data from web surveys. The paper presents five theorems and vector matching issues how to do this. Therefore, it is easy to apply them. The concept presented in the paper can be used to start the data combining on a large scale. However in the paper there is no such data combining.
Data sources	Web surveys, Phone interviews
Domains	Tourism, Population
Keywords	Data combining, Sample matching, Web surveys
Classification (A – very relevant, B – relevant, C – less relevant)	C

1.29. Chen H., Chiang, R. L., & Storey, V. C. (2012). Business Intelligence and Analytics: From Big Data to Big Impact. MIS Quarterly, 36(4), 1165-1188.

SPECIFICATION	DESCRIPTION

Bibliographic data	Chen H., Chiang, R. L., & Storey, V. C. (2012). Business Intelligence And Analytics: From Big Data To Big Impact. MIS Quarterly, 36(4), 1165-1188.
Link	https://misq.org/skin/frontend/default/misq/pdf/V36I4/SI_ChenIntroduction.pdf
Short overview (strengths, weaknesses)	Authors provide a systematic approach to define Big Data in terms of Business Intelligence 1.0, 2.0 and 3.0. They give five different fields of Big Data appliances: E-Commerce and Market Intelligence; E-Government and Politics 2.0; Science & Technology; Smart Health and Wellbeing; Security and Public Safety. They do not give a working solution to apply the use cases, just a theoretical approach to define it.
Data sources	Sensors, Web, Transactional data
Domains	Social statistics, Business statistics
Keywords	Big Data definition, Business Intelligence
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.30. Gang-Hoon, K., Trimi, S., & Ji-Hyong, C. (2014). Big-Data Applications in the Government Sector. Communications Of The ACM, 57(3).

SPECIFICATION	DESCRIPTION
Bibliographic data	Gang-Hoon, K., Trimi, S., & Ji-Hyong, C. (2014). Big-Data Applications in the Government Sector. Communications Of The ACM, 57(3), pp. 78-85.
Link	https://dl.acm.org/citation.cfm?id=2500873
Short overview (strengths, weaknesses)	The paper shows Big Data appliances in the government sector in East-Asia and Australia-Oceania regions. It describes the following data sources: web, biological and industrial sensors, video, email, and social communications to use them in the decision making process. The paper is mostly a discussion on new skills necessary (Hadoop, NoSQL) to process efficiently the data. The paper does not show a specific solution how to apply and integrate all data sources mentioned above.
Data sources	Sensors, Video, Email, Social media
Domains	Social statistics

Keywords	Government authorities, Decision making
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.31. Beyer M. A., Thoo E., Selvage M. Y. and Zaidi E. (2017): Magic Quadrant for Data Integration Tools. Gartner report, August version

SPECIFICATION	DESCRIPTION
Bibliographic data	Beyer M. A., Thoo E., Selvage M. Y. and Zaidi E. (2017): Magic Quadrant for Data Integration Tools. Gartner report, August version, ID Number: G00314940
Link	https://www.gartner.com/doc/3777464/magic-quadrant-data-integration-tools
Short overview (strengths, weaknesses)	Comparison and Evaluation of the integration tools available in the market in relation with how they incorporate legacy systems, innovation and transformational technologies and approaches.
Data sources	Not specified
Domains	Not specified
Keywords	Data integration, tools
Classification (A – very relevant, B – relevant, C – less relevant)	B

1.32. Singh D. and Reddy C.K. (2015): A Survey on Platforms for Big Data Analytics. Journal of Big Data, Springer 2015.

SPECIFICATION	DESCRIPTION
Bibliographic data	Singh D. and Reddy C.K. (2015): A Survey on Platforms for Big Data Analytics. Journal of Big Data, Springer 2015
Link	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4505391/
Short overview (strengths, weaknesses)	In-depth analysis of different platforms available for performing Big Data analytics based on the metrics: scalability, throughput, fault tolerance, data size, real-time processing and iterative task support.
Data sources	Not specified
Domains	Not specified
Keywords	Big Data Infrastructure

Classification (A – very relevant, B – relevant, C – less relevant)	B

1.33. Dufty D. et al., A Suggested Framework for the Quality of Big Data, Deliverables of the UNECE Big Data Quality Task Team, December, 2014.

SPECIFICATION	DESCRIPTION
Bibliographic data	Dufty D. et al., A Suggested Framework for the Quality of Big Data, Deliverables of the UNECE Big Data Quality Task Team, December, 2014.
Link	https://statswiki.unece.org/display/bigdata/2014+Project?preview=%2F108102944%2F108298642%2FBig+Data+Quality+Framework+-+final-+Jan08-2015.pdf
Short overview (strengths, weaknesses)	The paper is very important for the data quality of Big Data sources. It was written by official statistics representatives. Data quality can be evaluated based on three hyperdimensions, depending the entity being processed. It includes data, metadata and source. Each hyperdimension has different indicators listed. The paper itself is a base to develop other Big Data quality frameworks for different purposes of the paper.
Data sources	Administrative data, social media, unstructured datasets
Domains	Not specified
Keywords	Data quality, big data quality indicators
Classification (A – very relevant, B – relevant, C – less relevant)	A

2. Acronyms and abbreviations

- AAPOR - American Association for Opinion Research
- ACMS - Aircraft condition monitoring systems
- CDR - Call Detail Records
- CSV - Comma Separated Values
- ETL - Extraction, Transformation, Loading
- EU-SILC - European Union Statistics on Income and Living Conditions
- MCR - Mobile Call Records
- SVM - Support Vector Machines

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